

key

What's my Transformation?



We have spent time making a graph library of some parent functions. But what about all those other more complicated functions? Today we will explore how many functions are simply transformations of the parent function using Desmos. Go to student.desmos.com and type in the class code.

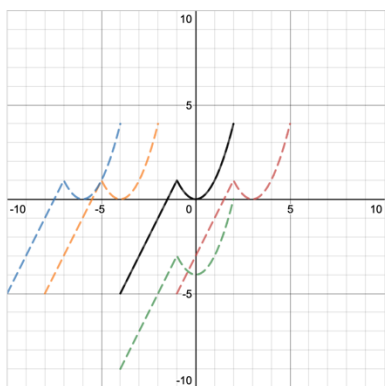
1. Why is there really only one parabola in the world (slide 2)?

All parabolas are just shifts + stretches of $y = x^2$

2. What happened to the parabola when you typed in $f(x - 4)$ (slide 6)?

It moved 4 to the right

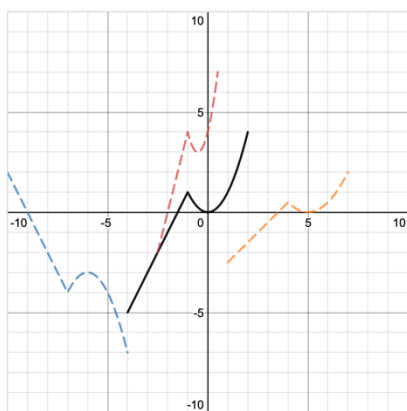
3. Record your answers from slide 7. Use colored pencils to trace each of the parabolas so they match the ones on Desmos.



Graph	Function
Red	$f(x-3)$
Orange	$f(x+4)$
Green	$f(x) - 4$ vertical shift
Blue	$f(x+b)$

horizontal shift

4. Starting in slide 9, the parabolas don't just shift around the page, they're actually getting wider and narrower! Record your answers from slide 13. Use colored pencils to trace the parabolas.



Graph	Function
Red	$f(2x+1) + 3$
Orange	$0.5 f(x-5)$
Blue	$-f(x+b) - 3$

horizontal dilation
vertical dilation
reflection

5. Based on what you learned in the Desmos activity, answer the following questions:

a. What's the difference between the graphs of $y = x^2 - 4$ and $y = (x - 4)^2$?

shift down

shift right 4

b. What's the difference between the graphs of $y = 3x^2$ and $y = \frac{1}{3}x^2$?

vertical stretch

vertical shrink

Lesson 1.5 – Transformations of Functions

translation

Transformation	Function Notation	Characteristics
Vertical Shift	$f(x) + c$	Moves up when... $c > 0$ Moves down when... $c < 0$
Horizontal Shift	$f(x + c)$	Moves right when... $c < 0$ Moves left when... $c > 0$
Vertical Dilations	$c f(x)$	Stretches in the y-direction when... $c > 1$ Shrinks/compresses in the y-direction when... $0 < c < 1$
Horizontal Dilations	$f(cx)$	Shrinks/compresses in the x-direction when... $c > 1$ Stretches in the x-direction when... $0 < c < 1$
Reflection about x-axis	$-f(x)$	$(x, y) \rightarrow (x, -y)$
Reflection about y-axis	$f(-x)$	$(x, y) \rightarrow (-x, y)$

Check Your Understanding

1. List the parent function and then the transformations that occurred to the parent function.

a. $f(x) = 2x^2 - 3$

b. $g(x) = |x - 4|$

c. $h(x) = -(4x)^3$

x^2
vertical stretch of 2
shift down 3

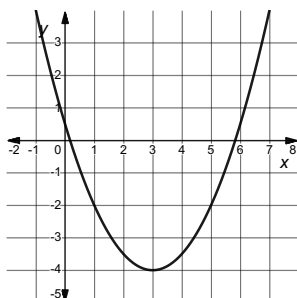
$|x|$
shift right 4

x^3
horizontal shrink by factor of $\frac{1}{4}$
reflection over x-axis

2. Give the equation of the parent function $y = \sqrt{x}$ after it has been shifted left 7 and up 2.

$$y = \sqrt{x+7} + 2$$

3. (Multiple Choice) The graph of g below is a transformation of a parent function. Which equation correctly describes $g(x)$?



- ☐ A) $g(x) = (x + 3)^2 - 4$
☐ B) $g(x) = 2(x + 3)^2 - 4$
☒ C) $g(x) = \frac{1}{2}(x - 3)^2 - 4$
☐ D) $g(x) = \left(\frac{1}{2}x - 3\right)^2 - 4$

4. (Multiple Choice) Pilar's parents give him an allowance amount $a(t)$, based on his age, t , in years. Which of the following expressions would give the allowance amount of Pilar's younger sister, who is three years younger than he is?

A) $a(t) - 3$

B) $a(t) + 3$

☒ C) $a(t - 3)$

D) $a(t + 3)$